

The Jacobian and Gradients of Matrices

Contents

1. Overview	2
2. Functions with several outputs	3
3. The Jacobian	3
3.1 Collecting all partial derivatives	3
3.2 One picture of every shape	4
4. The Jacobian as the best local linear map	5
4.1 Small steps are transformed by J	5
4.2 For a linear function, the Jacobian is its matrix	6
5. The Jacobian determinant: how areas scale	7
6. The chain rule with Jacobians	7
7. Example: the gradient of the least-squares loss	8
8. Gradients of matrices	9
9. Useful identities	10
10. Preview: backpropagation and automatic differentiation	11
11. Summary	13
12. Sources	13
13. Key terms	13

Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Linear transformations and matrices ([Note 500](#)); Matrix multiplication as composition ([Note 510](#)); Partial derivatives and gradients ([Note 601](#)).
- **Leads to:** Memoization ([Note 1019](#)).

1. Overview

Key point: For a function with several outputs, the derivative is a matrix, the Jacobian: one row per output, one column per input. Near a point, the function acts like the linear map of that matrix.

This Note follows Chapter 5 (Sections 5.3 to 5.6) of *Mathematics for Machine Learning* (Deisenroth, Faisal and Ong, 2020).

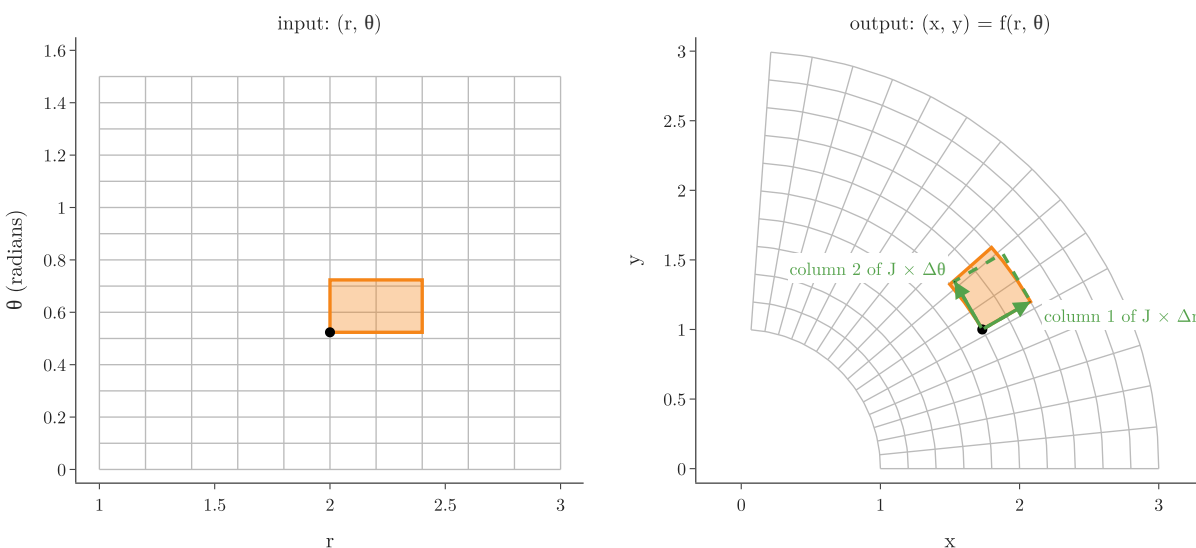


Figure 1: Polar coordinates bend a straight grid (left) into circles and rays (right). A small orange cell lands on a curved cell that is almost the green parallelogram given by the Jacobian

Figure 1 shows the main idea. A function of several outputs moves points of the plane, like the linear transformations of the [linear transformations and matrices Note](#). But it bends the grid. Zoom in on one small cell, though, and the bending almost disappears: the cell lands on something very close to a parallelogram. That parallelogram is drawn by a matrix, the Jacobian.

The [partial derivatives and gradients Note](#) handled functions with one output. Here we allow many outputs (Sections 2 to 6), apply this to the least-squares loss (Section 7), take a short

look at derivatives with respect to matrices (Sections 8 and 9), and preview how deep learning libraries compute gradients (Section 10).

2. Functions with several outputs

Key point: $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a stack of m ordinary functions $f_1, \dots, f_m$, each with n inputs.

A **vector-valued function** takes n numbers in and gives m numbers out. We can always read it as m separate functions stacked on top of each other:

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{bmatrix} \in \mathbb{R}^m, \quad f_i : \mathbb{R}^n \rightarrow \mathbb{R}$$

Each f_i has its own gradient, computed exactly as in the previous Note. Our running example is **polar coordinates**: a radius r and an angle θ go in, a point (x, y) of the plane comes out.

$$\mathbf{f}(r, \theta) = \begin{bmatrix} r \cos \theta \\ r \sin \theta \end{bmatrix}$$

At $r = 2$, $\theta = \pi/6$ (30°), the output is $(2 \times 0.866, 2 \times 0.5) = (1.732, 1)$, the black dot of Figure 1.

ML is full of such functions: a neural network layer turns a vector of inputs into a vector of outputs, and so does a prediction for all **observations** (records, one row of the data table each) of a dataset at once.

3. The Jacobian

Key point: The Jacobian is the $m \times n$ matrix of all first partial derivatives: row i is the gradient of output f_i , column j says how all outputs respond to input x_j .

3.1 Collecting all partial derivatives

Key point: Put the gradients of $f_1, \dots, f_m$ as rows on top of each other.

The partial derivative of $\mathbf{f}$ with respect to one input x_j is a column: how each of the m outputs responds to a nudge in x_j . Placing these n columns side by side gives a matrix.

1. **In words:** entry (i, j) is the partial derivative of output i with respect to input j . Row i is the gradient of f_i .

2. **Formula:** the **Jacobian** is

$$J = \frac{d\mathbf{f}}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} \in \mathbb{R}^{m \times n}, \quad J_{ij} = \frac{\partial f_i}{\partial x_j}$$

3. **Example:** for polar coordinates, differentiate each output with respect to r and to θ :

$$J = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}, \quad J(2, \frac{\pi}{6}) = \begin{bmatrix} 0.866 & -1 \\ 0.5 & 1.732 \end{bmatrix}$$

This arrangement, outputs as rows and inputs as columns, is the **numerator layout**. Some texts use the transpose (the denominator layout); the numbers are the same, only flipped.

3.2 One picture of every shape

Key point: Rows follow the outputs, columns follow the inputs; the derivative, the gradient and the Jacobian are all the same idea in different shapes.

The gradient of the previous Note is the special case of one output: a Jacobian with one row. Figure 2 sorts all four cases.

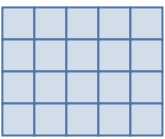
	input $x \in \mathbb{R}$	input $\mathbf{x} \in \mathbb{R}^D$
output $f \in \mathbb{R}$	 number (1×1) derivative $f'(x)$	 row ($1 \times D$) gradient ∇f
output $\mathbf{f} \in \mathbb{R}^E$	 column ($E \times 1$)	 matrix ($E \times D$) Jacobian J

Figure 2: The shape of the derivative $d\mathbf{f}/d\mathbf{x}$: one row per output, one column per input

Before computing any derivative, we write down its shape: a chain-rule product whose shapes do not fit is certainly wrong.

4. The Jacobian as the best local linear map

Key point: Near a point $\mathbf{x}_0$, $\mathbf{f}(\mathbf{x}_0 + \boldsymbol{\delta}) \approx \mathbf{f}(\mathbf{x}_0) + J\boldsymbol{\delta}$: the function acts like the linear transformation J on small steps.

4.1 Small steps are transformed by J

Key point: A small step $\boldsymbol{\delta}$ in the input becomes, almost exactly, the step $J\boldsymbol{\delta}$ in the output.

The [linear transformations and matrices Note](#) read a matrix by its columns: where each basis vector lands. The Jacobian's columns say the same thing for small steps: column j is where a small step along input j lands, per unit of step.

1. **In words:** start from the output at $\mathbf{x}_0$ and add the Jacobian times the step.
2. **Formula:**

$$\mathbf{f}(\mathbf{x}_0 + \boldsymbol{\delta}) \approx \mathbf{f}(\mathbf{x}_0) + J(\mathbf{x}_0) \boldsymbol{\delta}$$

3. **Example:** polar coordinates at $(2, \pi/6)$, step $\boldsymbol{\delta} = (0.1, 0.05)$:

$$\begin{bmatrix} 1.732 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.866 & -1 \\ 0.5 & 1.732 \end{bmatrix} \begin{bmatrix} 0.1 \\ 0.05 \end{bmatrix} = \begin{bmatrix} 1.732 + 0.037 \\ 1 + 0.137 \end{bmatrix} = \begin{bmatrix} 1.769 \\ 1.137 \end{bmatrix}$$

The exact value $\mathbf{f}(2.1, \pi/6 + 0.05)$ is $(1.764, 1.140)$: off by only 0.005.

The linear approximation is the tangent line of the [derivatives of one variable Note](#), in several dimensions. Figure 1 shows it as a picture: the orange cell lands on a curved cell, and the green parallelogram spanned by the Jacobian's columns (each times the cell's side) almost covers it. The smaller the cell, the better the match.

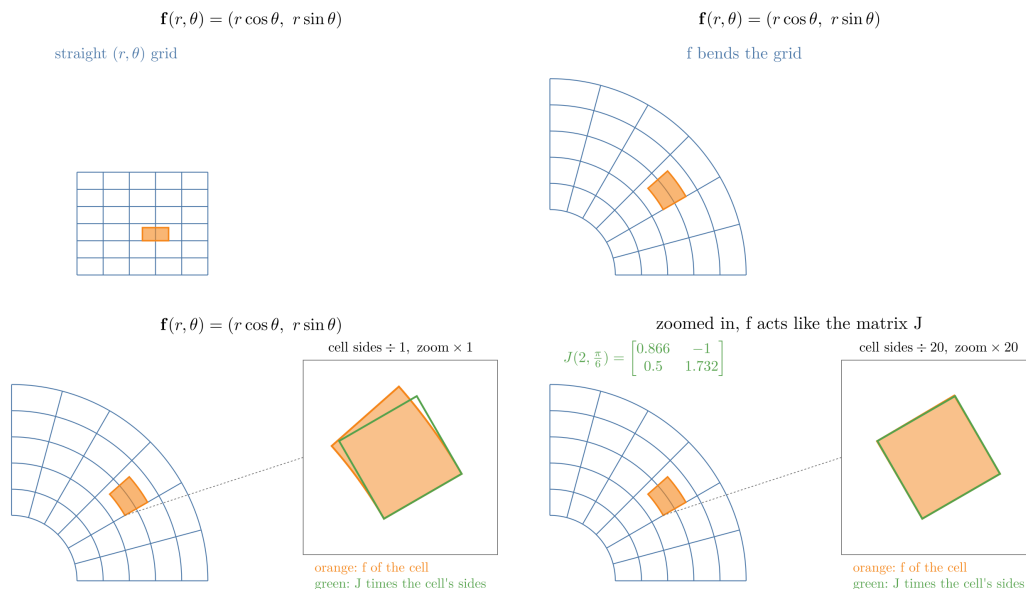


Figure 3: The polar map bends a straight (r, θ) grid into rays and arcs. Then we zoom on the orange cell at $(2, \pi/6)$: as the cell shrinks (shown magnified to a fixed size), its curved image (orange) becomes the parallelogram spanned by the Jacobian’s columns times the cell’s sides (green)

Figure 3 animates the same idea. Watch the gap between the orange and green outlines: at full size the curved cell sticks out, but with sides 20 times smaller the two outlines coincide. Zoomed in far enough, the function is the linear map J . The zoom-in picture of local linearity follows Sanderson’s Khan Academy lessons on the Jacobian.

4.2 For a linear function, the Jacobian is its matrix

Key point: The Jacobian of $\mathbf{f}(\mathbf{x}) = A\mathbf{x}$ is A itself, at every point.

1. **In words:** output i is $f_i = A_{i1}x_1 + \cdots + A_{in}x_n$, so its partial derivative with respect to x_j is just the number A_{ij} .
2. **Formula:** for $A \in \mathbb{R}^{m \times n}$,

$$\frac{d}{d\mathbf{x}} A\mathbf{x} = A$$

3. **Example:** for the matrix of the [linear transformations and matrices Note](#), A with rows $[1, 3]$ and $[-2, 0]$, we have $f_1 = x_1 + 3x_2$ and $f_2 = -2x_1$, so

$$J = \begin{bmatrix} 1 & 3 \\ -2 & 0 \end{bmatrix} = A$$

A linear map needs no approximation: the “best local linear map” is the map itself. The statement is the matrix version of “the derivative of ax is a ”.

5. The Jacobian determinant: how areas scale

Key point: $|\det J|$ is the factor by which the function stretches small areas (or volumes) near a point.

The [eigenvectors and eigenvalues Note](#) introduced the **determinant**: the factor by which a linear transformation scales areas. Since the Jacobian is the local linear map, its determinant tells how much $\mathbf{f}$ scales small areas near a point.

1. **In words:** a tiny region of area ΔA around $\mathbf{x}_0$ lands on a region of area about $|\det J(\mathbf{x}_0)| \Delta A$.
2. **Formula:** the **Jacobian determinant** for polar coordinates is

$$\det J = \cos \theta \cdot r \cos \theta - (-r \sin \theta) \sin \theta = r(\cos^2 \theta + \sin^2 \theta) = r$$

3. **Example:** the orange cell of Figure 1 has sides $\Delta r = 0.4$ and $\Delta \theta = 0.2$, area 0.08 in the input. The cell sits around $r \approx 2.2$, so it lands on an area of about $2.2 \times 0.08 = 0.176$ (here the exact area is also 0.176).

So polar coordinates stretch cells more the further they are from the origin: in Figure 1, cells on the outer arcs are bigger. For a linear map, $\det J = \det A$ everywhere: A above has $\det A = 1 \cdot 0 - 3 \cdot (-2) = 6$, so every area grows 6 times.

Extra: The Jacobian determinant is how probability densities change under a change of variables. If $\mathbf{y} = \mathbf{f}(\mathbf{x})$, probability in a small region must be kept, so the density of $\mathbf{y}$ is the density of $\mathbf{x}$ divided by $|\det J|$: where $\mathbf{f}$ stretches area, the same probability is spread thinner (MML §6.7). Its one-dimensional form, “density is a derivative”, appears in the [PDF and continuous CDF Note](#). Generative models called normalizing flows are built on exactly this rule (Rezende and Mohamed 2015).

6. The chain rule with Jacobians

Key point: The derivative of a composition is the product of the Jacobians, in the same order as the functions: $J_{\mathbf{f} \circ \mathbf{g}} = J_{\mathbf{f}} J_{\mathbf{g}}$.

The [partial derivatives and gradients Note](#) (Section 6) wrote the multivariate chain rule as a row gradient times a matrix of inner derivatives. That matrix was a Jacobian, and the rule holds for any shapes.

1. **In words:** multiply the Jacobian of the outer function (at the inner output) by the Jacobian of the inner function. The inner sizes must match, like any matrix product.
2. **Formula:** for $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^k$ and $\mathbf{f} : \mathbb{R}^k \rightarrow \mathbb{R}^m$,

$$\underbrace{\frac{d\mathbf{f}(\mathbf{g}(\mathbf{x}))}{d\mathbf{x}}}_{m \times n} = \underbrace{\frac{\partial \mathbf{f}}{\partial \mathbf{g}}}_{m \times k} \underbrace{\frac{\partial \mathbf{g}}{\partial \mathbf{x}}}_{k \times n}$$

3. **Example:** $h(t) = f(\mathbf{g}(t))$ with $f(\mathbf{x}) = x_1 x_2^2$ and $\mathbf{g}(t) = [2t, t + 1]$. At $t = 1$, $\mathbf{x} = (2, 2)$.

Shapes: $\partial f / \partial \mathbf{x}$ is 1×2 , $\partial \mathbf{g} / \partial t$ is 2×1 .

$$\frac{dh}{dt} = \begin{bmatrix} x_2^2 & 2x_1x_2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 & 8 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 16$$

Directly: $h(t) = 2t(t+1)^2$ has $h'(t) = 2(t+1)^2 + 4t(t+1)$, which is $8 + 8 = 16$ at $t = 1$.

The Jacobian chain rule is the [matrix multiplication as composition Note](#) again, zoomed in: near a point each function is a linear map, and doing one linear map after another multiplies their matrices.

7. Example: the gradient of the least-squares loss

Key point: Splitting $L(\boldsymbol{\theta}) = \|\mathbf{y} - \Phi\boldsymbol{\theta}\|^2$ into an error vector and its squared length, the chain rule gives $\partial L / \partial \boldsymbol{\theta} = -2\mathbf{e}^\top \Phi$.

The [multiple linear regression maths Note](#) differentiated the squared error by expanding it and applying two rules of **matrix calculus**. The chain rule gives the same result in three short steps, and the steps scale to models that are too deep to expand.

The model is $\hat{\mathbf{y}} = \Phi\boldsymbol{\theta}$, with Φ the $N \times D$ data matrix (one row per observation) and $\boldsymbol{\theta}$ the D parameters. Define two functions:

$$\mathbf{e}(\boldsymbol{\theta}) = \mathbf{y} - \Phi\boldsymbol{\theta} \quad (N \text{ errors}), \quad L(\mathbf{e}) = \|\mathbf{e}\|^2 = \mathbf{e}^\top \mathbf{e} \quad (\text{one number})$$

1. **In words:** first find the shapes, then the two pieces, then multiply.
 - $\partial L / \partial \boldsymbol{\theta}$ will be $1 \times D$ (one output, D inputs).
 - $\partial L / \partial \mathbf{e} = 2\mathbf{e}^\top$, a $1 \times N$ row: the derivative of $e_1^2 + \dots + e_N^2$ with respect to each e_n is $2e_n$.
 - $\partial \mathbf{e} / \partial \boldsymbol{\theta} = -\Phi$, an $N \times D$ matrix: the Jacobian of a linear function (Section 4.2), with a minus sign.

2. **Formula:**

$$\frac{\partial L}{\partial \boldsymbol{\theta}} = \underbrace{\frac{\partial L}{\partial \mathbf{e}}}_{1 \times N} \underbrace{\frac{\partial \mathbf{e}}{\partial \boldsymbol{\theta}}}_{N \times D} = -2\mathbf{e}^\top \Phi = -2(\mathbf{y} - \Phi\boldsymbol{\theta})^\top \Phi$$

3. **Example:** the three points of the gradient checking example in the [partial derivatives and gradients Note](#), with a column of ones for the intercept, $\boldsymbol{\theta} = [b, m] = [0, 1]$:

$$\Phi = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}, \quad \mathbf{e} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \quad -2\mathbf{e}^\top \Phi = -2[5, 11] = [-10, -22]$$

These are $\partial L / \partial b = -10$ and $\partial L / \partial m = -22$, exactly the values found there by hand.

Transposed, $-2\Phi^\top(\mathbf{y} - \Phi\boldsymbol{\theta}) = 2\Phi^\top\Phi\boldsymbol{\theta} - 2\Phi^\top\mathbf{y}$, the column form of the [multiple linear regression maths Note](#). Setting it to zero gives the normal equation.

Python:

```
import numpy as np

Phi = np.array([[1.0, 1.0], [1.0, 2.0], [1.0, 3.0]])
y = np.array([2.0, 4.0, 5.0])
theta = np.array([0.0, 1.0])

e = y - Phi @ theta          # [1, 2, 2]
dL_de = 2 * e                # the 1 x N piece
de_dtheta = -Phi             # the N x D piece
dL_de @ de_dtheta            # [-10, -22]
```

8. Gradients of matrices

Key point: A derivative of a matrix, or with respect to a matrix, has more than two indices: a tensor. In practice we flatten the matrices into long vectors so the Jacobian stays an ordinary matrix.

Some models need derivatives with respect to a whole weight matrix, such as a neural network layer's W . Counting indices shows the shape:

- an $m \times n$ matrix differentiated with respect to a vector of length p gives an $m \times n \times p$ array;
- an $m \times n$ matrix differentiated with respect to a $p \times q$ matrix gives an $m \times n \times p \times q$ array, with entries $\partial A_{ij} / \partial B_{kl}$.

Arrays with more than two indices are **tensors** (see the [tensors Note](#)).

There are two equivalent ways to handle them:

- **Collect partial derivatives into a tensor.** Each partial derivative is a matrix; stack them along a new axis.
- **Flatten first.** Reshape each matrix into one long vector (an $m \times n$ matrix becomes a vector of mn numbers), compute an ordinary $mn \times pq$ Jacobian, and reshape back if needed. The chain rule is then plain matrix multiplication.

A small example: $\mathbf{f} = A\mathbf{x}$ with A of size 2×3 and $\mathbf{x} = [1, 2, 3]$, differentiated with respect to A . Since $f_1 = A_{11}x_1 + A_{12}x_2 + A_{13}x_3$, the derivative of f_1 with respect to the entries of A is $\mathbf{x}$ in the first row and zeros elsewhere:

$$\frac{\partial f_1}{\partial A} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}, \quad \frac{\partial f_2}{\partial A} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 2 & 3 \end{bmatrix}$$

Stacked, these form a $2 \times 2 \times 3$ tensor. Each output depends only on its own row of A ; this is why the gradient of a layer's weights is built from its inputs $\mathbf{x}$.

9. Useful identities

Key point: A handful of ready-made gradients cover most ML derivations, the same way a table of basic derivatives covers one-variable calculus.

Gradients in the row (numerator) layout, for a variable vector $\mathbf{x}$, a variable matrix X , and fixed $\mathbf{a}$, $\mathbf{b}$, B :

Expression	Gradient	One-variable analogue
$\mathbf{a}^\top \mathbf{x}$ or $\mathbf{x}^\top \mathbf{a}$	$\mathbf{a}^\top$	$(ax)' = a$
$A\mathbf{x}$	A	$(ax)' = a$
$\mathbf{x}^\top B\mathbf{x}$	$\mathbf{x}^\top (B + B^\top)$	$(bx^2)' = 2bx$
$\mathbf{a}^\top X\mathbf{b}$ (with respect to X)	$\mathbf{a}\mathbf{b}^\top$	$(axb)' = ab$
$(\mathbf{x} - A\mathbf{s})^\top W(\mathbf{x} - A\mathbf{s})$ (with respect to $\mathbf{s}$, W symmetric)	$-2(\mathbf{x} - A\mathbf{s})^\top W A$	weighted least squares

The third row deserves a worked check.

1. **In words:** a quadratic form behaves like bx^2 , but B and $B^\top$ both contribute; for symmetric B the gradient is $2\mathbf{x}^\top B$.
2. **Formula:**

$$\frac{\partial}{\partial \mathbf{x}} \mathbf{x}^\top B\mathbf{x} = \mathbf{x}^\top (B + B^\top)$$

3. **Example:** B with rows $[2, 1]$ and $[0, 3]$, at $\mathbf{x} = [1, 2]$. Then $B + B^\top$ has rows $[4, 1]$ and $[1, 6]$, so

$$\mathbf{x}^\top (B + B^\top) = [1 \cdot 4 + 2 \cdot 1, \quad 1 \cdot 1 + 2 \cdot 6] = [6, \quad 13]$$

Multiplied out, $\mathbf{x}^\top B\mathbf{x} = 2x_1^2 + x_1x_2 + 3x_2^2$ has partial derivatives $4x_1 + x_2 = 6$ and $x_1 + 6x_2 = 13$.

The bowl of the [partial derivatives and gradients Note](#), $x_1^2 + x_1x_2 + 2x_2^2$, is $\mathbf{x}^\top B\mathbf{x}$ with the symmetric B of rows $[1, 0.5]$ and $[0.5, 2]$; the rule gives $2\mathbf{x}^\top B = [2x_1 + x_2, \quad x_1 + 4x_2]$, its gradient. The table of the [multiple linear regression maths Note](#) holds the same rules written as columns.

Extra: Reference tables also list rules for traces, determinants and inverses of matrices that depend on X , for example

$$\frac{\partial}{\partial X} \det(X) = \det(X) (X^{-1})^\top, \quad \frac{\partial}{\partial X} \mathbf{a}^\top X^{-1} \mathbf{b} = -(X^{-1})^\top \mathbf{a}\mathbf{b}^\top (X^{-1})^\top$$

They appear when fitting covariance matrices, as in the maximum likelihood fit of a Gaussian (Bishop §2.3.4). The standard collection is *The Matrix Cookbook* (Petersen and Pedersen 2012).

10. Preview: backpropagation and automatic differentiation

Key point: A deep network is a long chain of functions, so its gradient is a long product of Jacobians; backpropagation computes that product from the loss backwards, reusing every intermediate result.

A neural network computes its output as a composition of layers, $\mathbf{f}_K(\cdots \mathbf{f}_2(\mathbf{f}_1(\mathbf{x})))$, each layer with its own weights. By Section 6, the gradient of the loss with respect to the weights of an early layer is a product of the Jacobians of all later layers. Writing that product as one formula quickly becomes enormous.

The practical method breaks the function into elementary steps, a **computation graph**, and applies the chain rule one step at a time. Figure 4 does this for $f(x) = x^2 + e^{x^2}$.

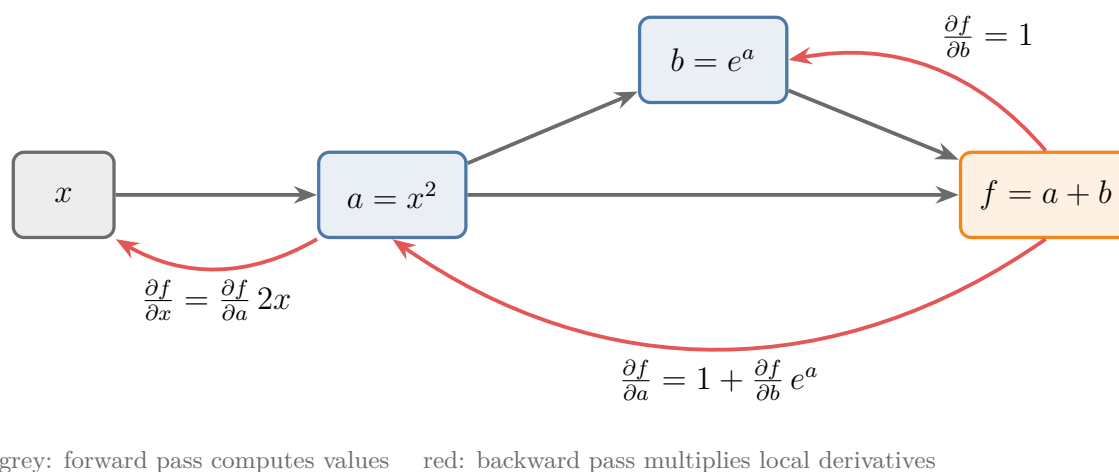


Figure 4: The computation graph of $f(x) = x^2 + e^{x^2}$: values flow forward (grey), derivatives flow backward (red)

1. **In words:** compute and keep every intermediate value going forward; then, starting from $\partial f / \partial f = 1$, multiply by each step's local derivative going backward, adding where two paths meet.
2. **Formula:** with $a = x^2$ and $b = e^a$,

$$\frac{\partial f}{\partial b} = 1, \quad \frac{\partial f}{\partial a} = 1 + \frac{\partial f}{\partial b} e^a, \quad \frac{\partial f}{\partial x} = \frac{\partial f}{\partial a} 2x$$

3. **Example:** at $x = 1$: forward, $a = 1$, $b = e = 2.718$, $f = 3.718$. Backward, $\partial f / \partial b = 1$, $\partial f / \partial a = 1 + 2.718 = 3.718$, $\partial f / \partial x = 3.718 \times 2 = 7.437$. The formula $f'(x) = 2x + 2x e^{x^2}$ gives $2 + 2e = 7.437$.

The backward pass has one step per forward step, each a multiplication by a local derivative, so computing the gradient takes work of the same order as computing f itself (MML §5.6). This backward pass is **backpropagation**; done automatically by software for any program,

it is **automatic differentiation** (reverse mode). Backpropagation is taught in full with neural networks, in the Deep Learning Notes.

Extra: Automatic differentiation is neither symbolic differentiation (which writes out a formula for f') nor a numerical difference quotient (which only estimates it). Automatic differentiation gives the exact derivative, up to rounding, by applying the chain rule to the actual operations a program runs. The method has a **forward mode**, which multiplies the Jacobians from the input side, and a **reverse mode**, which starts from the output. With far more inputs (weights) than outputs (the loss), the reverse mode is significantly cheaper (MML §5.6; Baydin et al. 2018), which is why deep learning libraries use it for training.

11. Summary

Object	Shape	Example
Jacobian $J = d\mathbf{f}/d\mathbf{x}$ of $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$	$m \times n$	polar at $(2, \pi/6)$: rows $[0.866, -1]$, $[0.5, 1.732]$
Jacobian of $A\mathbf{x}$	$m \times n$	A
Jacobian determinant	number	polar: r ; linear: $\det A$
Chain rule $J_{\mathbf{f} \circ \mathbf{g}} = J_{\mathbf{f}} J_{\mathbf{g}}$	$(m \times k)(k \times n)$	$[4, 8] \cdot [2, 1]^T = 16$
Least-squares gradient	$1 \times D$	$-2\mathbf{e}^T \Phi = [-10, -22]$
Gradient with respect to a matrix	tensor	flatten to keep it a matrix

- A vector-valued function is a stack of ordinary functions; its Jacobian stacks their gradients as rows.
- Near a point, the function acts like the linear map J : $\mathbf{f}(\mathbf{x}_0 + \boldsymbol{\delta}) \approx \mathbf{f}(\mathbf{x}_0) + J\boldsymbol{\delta}$.
- $|\det J|$ is the local area (volume) scaling factor.
- The chain rule multiplies Jacobians; checking shapes first prevents most mistakes.
- Backpropagation applies this chain rule backward through a computation graph.

12. Sources

Built from

- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Sections 5.3–5.6 and 6.7 (MML).
- Sanderson, G. (Khan Academy). “Local linearity for a multivariable function” and “The Jacobian matrix”, *Multivariable calculus*, lesson [“Jacobian”](#).

Other references

- Baydin, A. G., Pearlmutter, B. A., Radul, A. A. and Siskind, J. M. (2018). “Automatic Differentiation in Machine Learning: a Survey”. *Journal of Machine Learning Research* 18(153).
- Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer. Section 2.3.4, maximum likelihood for the Gaussian.
- Petersen, K. B. and Pedersen, M. S. (2012). *The Matrix Cookbook*. Technical University of Denmark.
- Rezende, D. J. and Mohamed, S. (2015). “Variational Inference with Normalizing Flows”. *ICML*.

13. Key terms

Term	Meaning
Vector-valued function	$\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$: several numbers in, several numbers out; a stack of m ordinary functions
Jacobian	The $m \times n$ matrix of all first partial derivatives, $J_{ij} = \partial f_i / \partial x_j$
Numerator layout	Writing derivatives with outputs as rows and inputs as columns
Polar coordinates	Describing a point by its distance r from the origin and its angle θ : $(r \cos \theta, r \sin \theta)$
Local linear map	The linear transformation a smooth function behaves like near a point; its matrix is the Jacobian
Jacobian determinant	$\det J$: the factor by which a function scales small areas or volumes near a point
Chain rule with Jacobians	The Jacobian of a composition is the product of the Jacobians, in the same order
Least-squares loss	$\ \mathbf{y} - \Phi\boldsymbol{\theta}\ ^2$; its gradient is $-2(\mathbf{y} - \Phi\boldsymbol{\theta})^\top \Phi$
Flattening	Reshaping a matrix into one long vector so that derivatives stay matrices
Quadratic form	$\mathbf{x}^\top B \mathbf{x}$; its gradient is $\mathbf{x}^\top (B + B^\top)$
Computation graph	A function broken into elementary steps, each a node, with arrows for the flow of values
Backpropagation	Computing the gradient of a loss by applying the chain rule backward through the computation graph
Automatic differentiation	Software that computes exact derivatives of a program by the chain rule on its elementary operations; reverse mode is backpropagation